

**BEFORE YOU READ — ANSWER FROM MEMORY**

1. Metronidazole kills trophozoites but not this other form of *E. histolytica* — which one?
2. Name the three luminal-only amebicides that never touch tissue disease.
3. Which drug is the DOC for onchocerciasis, and what does it fail to kill?

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 Intestinal protozoan infections — Amebiasis

Clinical form	What happens
Asymptomatic intestinal infection	Excretes infectious cysts — a source for further spread
Colitis	Trophozoites invade colonic mucosa → bloody diarrhoea (amebic dysentery)
Extraintestinal	Trophozoites cross into the portal circulation → amebic liver abscess

2 Metronidazole & tinidazole — the cornerstone

Kills	Trophozoites only — not cysts
Spectrum	Protozoa: <i>E. histolytica</i> , <i>Giardia lamblia</i> , <i>Trichomonas vaginalis</i> . Anaerobic bacteria: <i>C. difficile</i> , <i>C. perfringens</i> , <i>H. pylori</i> , <i>Fusobacterium</i> , <i>Bacteroides fragilis</i> , <i>Gardnerella vaginalis</i>
Mechanism	Prodrug → nitroreductase in the organism reduces the nitro group → cytotoxic metabolite → binds DNA → DNA fragmentation, inhibits synthesis. Tinidazole works the same way.
Pharmacokinetics	Rapid oral absorption, peak 1–3h. Low protein binding (10–20%). Half-life ~8h (metronidazole), 12–14h (tinidazole). Distributes into CSF, breast milk, vaginal/seminal fluid. Hepatic (CYP450) metabolism + glucuronidation; renal excretion — reduce dose in liver impairment.
Adverse effects	Nausea, metallic taste, dry mouth, rash, dysuria, dark urine; rare neurotoxicity (dizziness, vertigo, paraesthesia). Disulfiram-like reaction with alcohol (nausea, flushing, tachycardia, hypotension). Tinidazole: similar profile, better tolerated.
Interactions	Phenobarbitone/rifampin (enzyme inducers) ↓ effect. Cimetidine (enzyme inhibitor) ↑ toxicity. ↑ warfarin effect. ↓ renal lithium clearance → toxicity risk.
Indications	Amebiasis, giardiasis, trichomoniasis, cutaneous leishmaniasis; sepsis/peritonitis/meningitis/brain abscess (IV, sensitive organisms); pseudomembranous colitis; <i>H. pylori</i> PUD; surgical anaerobic prophylaxis; skin/soft tissue and periodontal infections (topical).

Mnemonic — metronidazole's non-anaerobe spectrum: “**GET GAP**” — **G**iardia, **E**ntamoeba, **T**richomonas, **G**ardnerella, **A**naerobes, **P**ylori.

3 The rest of the amebicide toolkit

Drug	Activity	Use
Iodoquinol	Luminal trophozoites + cysts only	Combined with metronidazole for amebic infection
Paromomycin	Luminal <i>E. histolytica</i> only (not absorbed from gut)	Also an alternative for cryptosporidiosis and giardiasis; safe choice in pregnancy
Diloxanide furoate	Luminal amebicide, not active on tissue trophozoites	Drug of choice for asymptomatic luminal infection
Chloroquine	Kills trophozoites in the liver	Combined with metronidazole for amebic liver abscess; no use in luminal disease

Emetine / dehydroemetine	Active against tissue trophozoites	Limited use — major toxicity concerns
Nitazoxanide	Inhibits anaerobic energy metabolism enzyme	Cryptosporidiosis, giardiasis, amebiasis; also some intestinal helminths

Mnemonic — Iodoquinol, Paromomycin, Diloxanide: “PID stays in the lumen” — none of the three touch tissue disease.

Giardiasis & cryptosporidiosis, in brief

Key points	
Giardiasis	Ingested cysts from contaminated food/water. Metronidazole 5–7 days usually works; a single dose of tinidazole may be superior. Paromomycin in pregnancy.
Cryptosporidiosis	Coccidian parasite, faecal-oral or waterborne. Self-limited if immunocompetent; can require hospitalisation in AIDS/immunocompromised. Nitazoxanide has shown activity in immunocompetent patients.

4 Intestinal helminthic infestations

Group	Examples
Nematodes (roundworms)	Roundworm, hookworm, whipworm, pinworm
Trematodes (flukes)	Schistosoma, Fasciola, Clonorchis, Opisthorchis, Paragonimus
Cestodes (tapeworms)	T. saginata, T. solium, H. nana, D. latum

Drugs for nematodes

Benzimidazoles (albendazole, mebendazole)	MOA: bind parasite β -tubulin → block microtubule polymerisation, and irreversibly block glucose uptake → worm is expelled in the faeces. First-line for whipworm, pinworm, hookworm, and roundworm. ADR: abdominal pain, diarrhoea. Avoid in pregnancy.
Pyrantel pamoate	Poorly absorbed — acts locally in the gut. Depolarising neuromuscular blocker: releases acetylcholine and inhibits cholinesterase → spastic paralysis of the worm, which is then expelled. Effective for roundworm, pinworm, hookworm.
Ivermectin	Drug of choice for cutaneous larva migrans, strongyloidiasis, and onchocerciasis (river blindness) — kills microfilariae but not adult worms . More effective in ascariasis/enterobiasis than in trichuriasis or hookworm. MOA: opens glutamate-activated Cl^- channels in nematode nerve/muscle → hyperpolarisation → paralysis and death. Oral, does not readily cross the blood-brain barrier. Avoid in pregnancy.

Mnemonic — “IVERmectin for the RIVER (blindness)”; benzimidazoles “BEND” the worm's tubulin.

Drugs for trematodes (flukes)

Praziquantel	Effective against all schistosome species and most other trematode and cestode infections. MOA: increases trematode/cestode membrane permeability to Ca^{++} → paralysis, dislodgement, death. PK: ~80% oral bioavailability, take with food (bitter taste, don't chew), peak 1–3h, ~80% protein bound, penetrates CSF. ADR: dizziness, malaise, headache, GI upset. Interactions: CYP inducers (phenytoin, carbamazepine, phenobarbital) and corticosteroids ↓ levels; cimetidine ↑ levels. Contraindicated in ocular cysticercosis — killing the organism in the eye can cause irreversible damage.
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Drugs for cestodes (tapeworms)

Drug	Use
Albendazole	Primary use here is cysticercosis and hydatid disease; also effective for microsporidiosis
Niclosamide	Alternative to praziquantel for taeniasis, diphyllorhynchiasis, and other tapeworm infections

P Must-remember pearls

- Metronidazole kills trophozoites, not cysts — pair it with a luminal amebicide for a definitive cure.
- Iodoquinol, Paromomycin, and Diloxanide furoate are luminal-only — none of them treat tissue disease.
- Benzimidazoles are first-line for whipworm, pinworm, hookworm, and roundworm; ivermectin owns strongyloidiasis and onchocerciasis instead.
- Praziquantel raises Ca²⁺ permeability in worm membranes — and must never be used in ocular cysticercosis.
- A disulfiram-like reaction with alcohol is classic for metronidazole and tinidazole.

Q Test yourself

1. A patient is found to be an asymptomatic *E. histolytica* cyst passer with no colitis or liver involvement. Best drug?

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| A) Metronidazole alone | B) Diloxanide furoate |
| C) Chloroquine | D) Emetine |

2. A patient with an amebic liver abscess is treated with metronidazole plus a second drug that clears trophozoites from the liver. What is it?

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| A) Chloroquine | B) Paromomycin |
| C) Niclosamide | D) Pyrantel pamoate |

3. A pregnant woman is diagnosed with giardiasis. Which drug avoids the mutagenicity concerns of the usual first-line agents?

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| A) Tinidazole | B) Metronidazole |
| C) Paromomycin | D) Nitazoxanide |

4. A patient with neurocysticercosis involving the eye is being considered for praziquantel. Why is this contraindicated?

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| A) It doesn't cross the blood-brain barrier | B) It can cause irreversible ocular damage as the organism dies |
| C) It has no activity against cestodes | D) It is teratogenic |

5. A patient treated with ivermectin for onchocerciasis still has adult worms on follow-up. Why?

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| A) Ivermectin kills microfilariae, not adult worms | B) The patient needs a benzimidazole instead |
| C) Ivermectin only works on nematode eggs | D) Resistance is universal |

ANSWERS — COVER UNTIL YOU'VE COMMITTED

- 1** — B — Diloxanide furoate, the drug of choice for asymptomatic luminal infection.
2 — A — Chloroquine, used in combination with metronidazole for amebic liver abscess.
3 — C — Paromomycin, used in pregnancy to avoid possible mutagenic effects of the other agents.
4 — B — destroying the organism in the eye can trigger an inflammatory response causing irreversible damage.
5 — A — ivermectin kills microfilariae but has no activity against adult *Onchocerca volvulus*.

SPACED REVIEW

☐ Day 1

☐ Day 3

☐ Day 7

☐ Day 21

☐ Pre-exam

Pharmacology GIT Block · MEDucation by Mattin Majid · Source: Dr. Begard O. Berzinji lecture slides, LG 2.
Study alongside the lecture, not instead of it.